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Please write clearly in block capitals.

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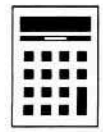
Functional Skills Level 2
MATHEMATICS

Paper 2 Calculator

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a calculator
- mathematical instruments.



Instructions
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
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For Examiner's Use	
Question	Mark
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Full Marks / Exemplary (100% Pass · Model Solutions)

· Assessment: November 2021 - Paper 2

· Subject: AQA Functional Skills Maths Level 2

· Complete accredited step-by-step models and tutor guidance

· Exam Board: AQA (Ofqual Regulated Reform Standard)

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Section A

Answer **all** questions in the spaces provided.

1 The probability of an event happening is $\frac{3}{5}$

Circle the probability of the event **not** happening.

[1 mark]

1 $\frac{2}{5}$ $\frac{3}{5}$ 0

2 $x = 5$ and $y = -3.2$

Work out the value of $x + y^2$

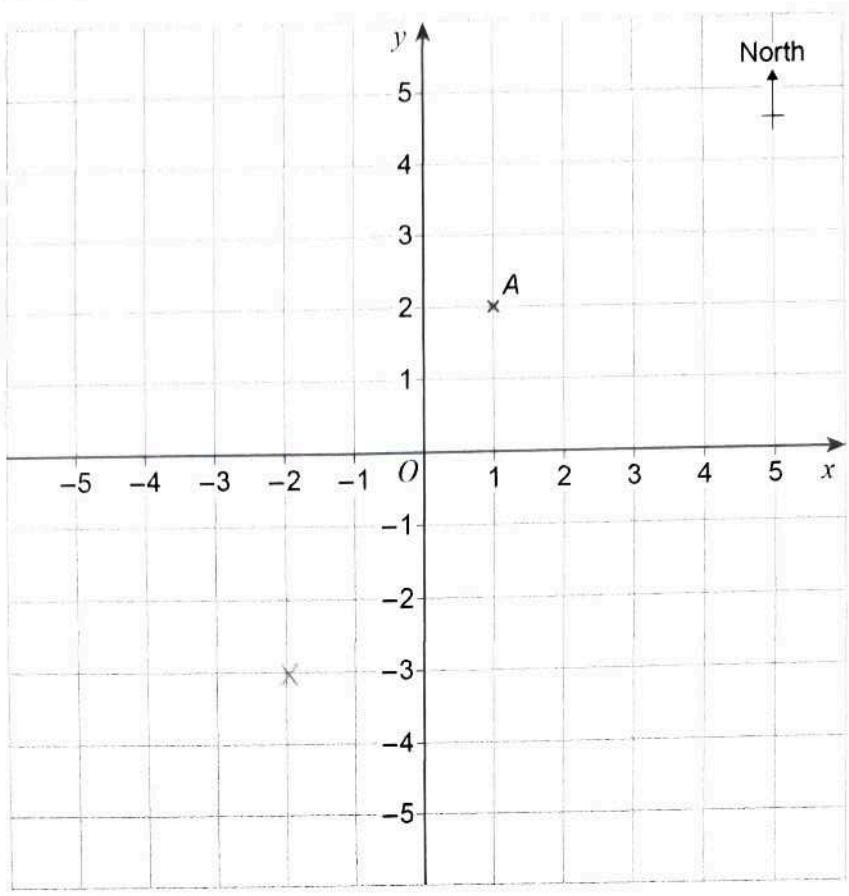
[2 marks]

$5 + (-3.2)^2 = 15.24$

Answer 15.24

outside the box

3



A is the point (1, 2)

B is the point

3 squares West of *A*

and

5 squares South of *A*.

Work out the coordinates of *B*.

[2 marks]

Answer (−2 , −3)

Turn over ▶

Do not write outside the box

4 In the boxes, write the number that is

3 more than −8 | −5 |

4 less than −2 | −6 |

halfway between −7 and 5 | −1 |

[3 marks]

5 Write the ratio 144 : 90 in its simplest form.

[2 marks]

144 : 90

16 : 10

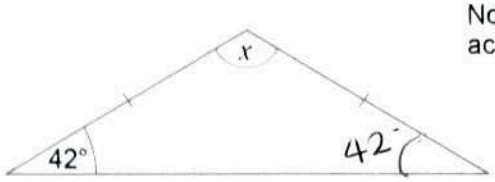
8 : 5

Answer 8 : 5

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6 Here is an isosceles triangle.

Not drawn accurately



Work out the size of angle x.

[2 marks]

180 - 2(42) = 96

Answer 96 °

12

Turn over for Section B

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

7 Energy

Jim's house has gas central heating.

7 (a) The charges for the gas Jim uses are shown in the table.

Cost per unit of gas (kWh)	2.92p
Standing charge per day	39.6p

In November, Jim uses 1976 kWh of gas.

Use **approximations** to estimate Jim's November gas payment.

Give your answer in pounds.

[5 marks]

1976 x 2.92 =

approximately → 2000 x 3 = 6000

per day approx 40 x 30 days

= 1200

Total = 6000 + 1200

= 7200p

= £72

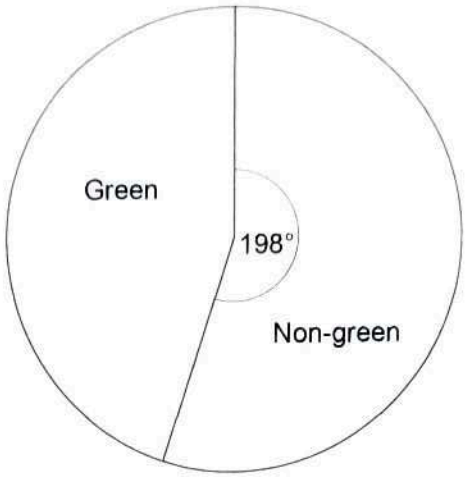
Answer £ 72

Do not write outside the box

7 (b) Jim's current gas company produces 43% of their gas from green sources.

He sees this pie chart for another gas company, MixGas.

Gas source



Jim wants to use the company with a greater proportion of gas from green sources.

Should Jim change to MixGas?

You **must** show your working.

[3 marks]

$360° - 198° = 162°$

proportion of total circle $\frac{162}{360} = 0.45 \times 100 = 45\%$

$45\% > 43\%$

Yes Jim should change to MixGas

Question 7 continues on the next page

Do not write outside the box

7 (c) Jim is thinking of buying solar panels.

The price of these panels has decreased by 27% since 2016

Jim would need to pay £4803.40 for the panels now.

How much would the panels have cost in 2016?

[3 marks]

100 - 27 = 73

4803.40 ÷ 73 x 100 = 6580

Answer £ 6580

11

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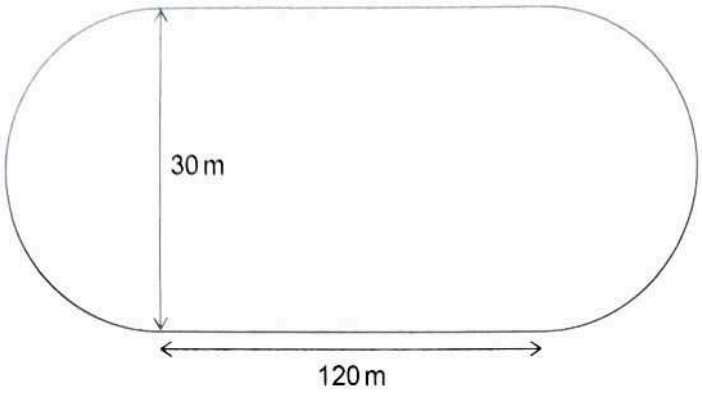
8 **Dog**

Ben has a dog called Pluto.

8 (a) Ben takes Pluto for a walk.

They walk **4 laps** of a sports track.

The sports track is the shape of two semi-circles joined by two straight sections.



Not drawn accurately

Ben has a target to walk 5000 m each day.

What percentage of his daily target has Ben completed on this walk?

[5 marks]

area of a hemisphere = $\frac{1}{2}\pi d$

$= \frac{1}{2} \pi (30) = 47.12$

2 hemispheres so $47.12 \times 2 = 94.25$

Total = $94.25 + 120 + 120 = 334.2$

4 laps = $334.2 \times 4 = 1337$

% = $1337 \div 5000 \times 100$

Answer 27 %

Question 8 continues on the next page

Do not write outside the box

8 (b) Pluto needs to eat 420 grams of food each day, in the ratio

wet food : dry food = 2 : 5

Ben buys a 12 kilogram bag of **dry** food.

How many days of **dry** food does the bag provide?

[4 marks]

420 grams 2 : 5

420 ÷ 7 = 60g

wet = 60 x 2 = 120g

dry = 60 x 5 = 300g

12kg = 12000g

12000 ÷ 300 = 40 days

Answer 40 days

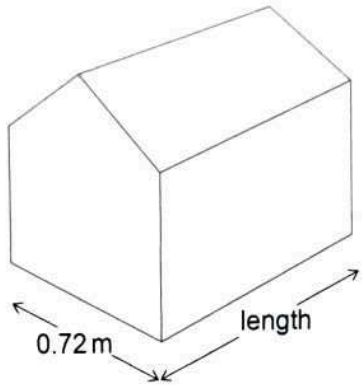
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8 (c) In his garden, Ben has a rectangular fenced area for Pluto.

The fenced area measures 2.4 m by 2.25 m

Ben builds a kennel for Pluto which covers $\frac{1}{6}$ of the fenced area.

Here is a sketch of the kennel.



The base of the kennel is a rectangle.

Work out the length of the kennel.

[4 marks]

Fenced area:
$2.4\text{m} \times 2.25\text{m} = 5.4\text{m}^2$

Kennel:
$\frac{1}{6}$ of $5.4\text{m}^2 = 0.9\text{m}^2$

$0.9\text{m}^2 = 0.72\text{m} \times \text{length}$

$\text{length} = \frac{0.9}{0.72}$

$= 1.25\text{m}$

Answer 1.25 m

13

Turn over for the next question

Do not write outside the box

9 Fundraising

A charity organises an annual sponsored swim.

9 (a) Ahmed swam 50 lengths of a pool that is 25 metres long.

The swim took him 38 minutes.

Ahmed says,

"My average speed was more than 0.7 metres per **second**."

Is Ahmed correct?

You **must** show your working.

[4 marks]

50 lengths x 25m = 1250 m

38 mins = 2280 seconds

speed in metres per second = $\frac{1250}{2280}$

= 0.548 ms^{-1}

No his speed was less than 0.7 ms^{-1}

Do not write outside the box

9 (b) Ahmed collected £35 for his swim and gave it to the charity.

$\frac{6}{7}$ of this money was donated by taxpayers.

Using Gift Aid, the charity claimed an extra 25p for every £1 donated by a taxpayer.

How much money did the charity receive, **including** Gift Aid, from Ahmed's swim?

[3 marks]

$35 \times \frac{6}{7} = 30$

£30 by taxpayers

25p extra per £1

£30 x 0.025 = 7.5

total = £35 + £7.5 = £42.50

Answer £ 42.50

Question 9 continues on the next page

Do not write outside the box

9 (c) This year, 20 people completed the sponsored swim.

The table shows information about the sponsor money collected.

Money (£x)	Frequency	Mid-point	frequency x midpoint
$0 < x \leqslant 10$	7	5	35
$10 < x \leqslant 20$	4	15	60
$20 < x \leqslant 30$	6	25	150
$30 < x \leqslant 40$	1	35	35
$40 < x \leqslant 50$	2	45	90
	Total = 20		370

Last year, the mean amount collected per swimmer was £14.85

The charity organiser says,

"This year's mean amount is **more than** £3 higher than last year's."

Show your working to support this statement.

[5 marks]

7 x 5 = 35

4 x 15 = 60

6 x 25 = 150

1 x 35 = 35

2 x 45 = 90

370

370 ÷ 20 = 18.5 mean

difference between this year and last year:

£18.5 - £14.85 = £3.65

∴ Yes, more than £3 higher

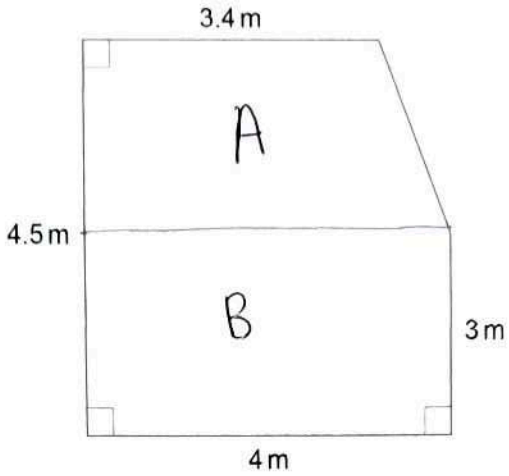
12

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10 **Conservatory**

Molly is building a conservatory.

10 (a) Here is a plan view of the conservatory.



Not drawn accurately

The conservatory floor will be made of concrete.

The concrete floor will be 0.1 m deep.

What volume of concrete will Molly need for the floor?

State the units of your answer.

[6 marks]

Area = area A + area B

A : $\frac{1}{2}(3.4 + 4) \times (4.5 - 3)$

$\frac{1}{2}(7.4) \times 1.5 = 5.55$

B : $3 \times 4 = 12$

Total area = $12 + 5.55 = 17.55$

volume = $17.55 \times 0.1 = 1.755 m^3$

Answer $1.755 m^3$

Question 10 continues on the next page

Do not write outside the box

10 (b) Molly has a scale drawing of the side elevation of the conservatory.

The scale of the drawing is 3 : 50

The height of the conservatory on the scale drawing is 23.4 centimetres.

Is the actual height of the conservatory **less than** 4 metres?

You **must** show your working.

[3 marks]

Scale height = 23.4cm

3 : 50

23.4 ÷ 3 = 7.8cm

7.8 x 50 = 390cm

390cm = 3.9m

Yes the actual height is less than 4m (3.9m).

Do not write outside the box

10 (c) Molly wants to build a brick wall as part of the conservatory.

It would take Molly 12 hours to build the wall by herself.

Molly pays 2 builders to work with her to build the wall.

Molly pays each builder £14.73 per hour.

How much will Molly pay each builder for their work?

Assume that Molly and the builders work at the same rate.

[3 marks]

12 hours ÷ 3 = 4 hours with 3 builders (Molly + 2)

4 x £14.73

= £58.92

Answer £ 58.92

12

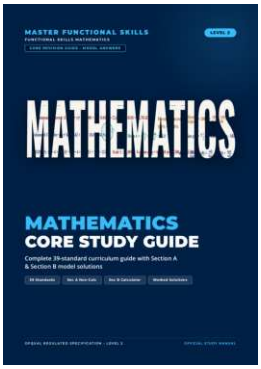
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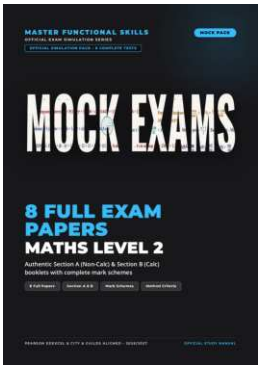


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